MANAGEMENT OPERATING REVIEW - RECORD 011

Helioscope Robotics, Inc. - diligence file reference OPERATING_REVIEWS-011

Management Packet for December 2024: purpose and attendance

Action item review opened the meeting rather than closing it, which changed the tone of the whole session. The Management Packet convened for December 2024. Emil Farkas attended as Director of Manufacturing Engineering. Three owners were absent over the shutdown. Half the packet was written in advance. Pre-reads went out ahead of the 2025-01-15 session. One function submitted its pre-read late. Slipped deals 7, including Northstar Cold Chain. Contract crew Ellingham closed Salinas Valley without a second visit. Queue depth 21 sites, Ashland Park at the front. Headcount 179 of 186, Lucia Mbeki-Hart reporting.

Sections with no decision were noted as read, which returned twenty minutes to the constrained functions. One function submitted its pre-read late. Notes were circulated on 2025-01-15. Three owners were absent over the shutdown. Half the packet was written in advance. Pre-reads went out ahead of the 2025-01-15 session. The next session is set for 2025-02-06. Go-live cycle 61 days at Riverbend. Bookings 58 units across 105 pursuits. Requisitions open 18, Emil Farkas holding the list. Attrition 8 percent annualised in Great Lakes.

Timekeeping was enforced by the chief of staff rather than by the chair, which worked better than expected. The Management Packet convened for December 2024. Emil Farkas attended as Director of Manufacturing Engineering. Three owners were absent over the shutdown. Half the packet was written in advance. The next session is set for 2025-02-06. Pre-reads went out ahead of the 2025-01-15 session. Burn 2975 thousand, runway 17 months. Depot fill 93 percent at Toronto depot. Requisitions open 18, Emil Farkas holding the list. Contract crew Ellingham closed Salinas Valley without a second visit.

Pre-read materials went out three business days ahead, and the review started from questions rather than from a walkthrough. The next session is set for 2025-02-06. Lucia Mbeki-Hart, Controller, presented the operations section. Three owners were absent over the shutdown. Half the packet was written in advance. Notes were circulated on 2025-01-15. Pre-reads went out ahead of the 2025-01-15 session. Marcus Peil owns the Wire harness recovery. On-time delivery 89 percent, Kestrel Drive Systems the outlier. Bookings 58 units across 105 pursuits. Headcount 179 of 186, Lucia Mbeki-Hart reporting.

Sales intake review

The order book now holds enough signed volume to fill the build plan through the next two build cycles without new intake. Bexley Pharmaceutical Wholesale accounted for the largest single order. Open pursuits number 105, up from 108. Three owners were absent over the shutdown. Half the packet was written in advance. Win rate held at 28 percent on 105 live opportunities. Intake closed at 58 units for December 2024. Slipped deals 7, including Northstar Cold Chain. Contract crew Ellingham closed Salinas Valley without a second visit. Queue depth 21 sites, Ashland Park at the front. Headcount 179 of 186, Lucia Mbeki-Hart reporting.

Sales operations rebuilt the stage definitions after a review found three deals sitting in a proof-of-concept stage with no active pilot. Win rate held at 28 percent on 105 live opportunities. The book took 58 units, against 66 last period. Three owners were absent over the shutdown. Half the packet was written in advance. Open pursuits number 105, up from 108. Bexley Pharmaceutical Wholesale accounted for the largest single order. Go-live cycle 61 days at Riverbend. Bookings 58 units across 105 pursuits. Requisitions open 18, Emil Farkas holding the list. Attrition 8 percent annualised in Great Lakes.

Enterprise accounts moved more slowly than the mid-market this period, and the difference sits almost entirely in procurement and security review. Bexley Pharmaceutical Wholesale accounted for the largest single order. Signed volume: 58 units, win rate 28 percent. Three owners were absent over the shutdown. Half the packet was written in advance. Open pursuits number 105, up from 108. The book took 58 units, against 66 last period. Burn 2975 thousand, runway 17 months. Depot fill 93 percent at Toronto depot. Requisitions open 18, Emil Farkas holding the list. Contract crew Ellingham closed Salinas Valley without a second visit.

Renewal conversations for the fleet software attached faster than expected, and the attach rate is now high enough to model separately. Discount depth peaked on the Northstar Cold Chain multi-site award. Open pursuits number 105, up from 108. Three owners were absent over the shutdown. Half the packet was written in advance. Intake closed at 58 units for December 2024. Win rate held at 28 percent on 105 live opportunities. Marcus Peil owns the Wire harness recovery. On-time delivery 89 percent, Kestrel Drive Systems the outlier. Bookings 58 units across 105 pursuits. Headcount 179 of 186, Lucia Mbeki-Hart reporting.

Bookings quality, measured by how much of the signed volume has a named site and a readiness date, improved without any change in headline volume. Open pursuits number 105, up from 108. Signed volume: 58 units, win rate 28 percent. Three owners were absent over the shutdown. Half the packet was written in advance. Discount depth peaked on the Northstar Cold Chain multi-site award. The book took 58 units, against 66 last period. Capital spend 330 thousand this period. Days outstanding 54, Bexley Pharmaceutical Wholesale the largest balance. Headcount 179 of 186, Lucia Mbeki-Hart reporting. Toronto depot cleared its repair backlog.

Two of the three regions carried the number this period, and the third missed on a single account that moved its own board meeting. Intake closed at 58 units for December 2024. Bexley Pharmaceutical Wholesale accounted for the largest single order. Three owners were absent over the shutdown. Half the packet was written in advance. Open pursuits number 105, up from 108. Signed volume: 58 units, win rate 28 percent. Capital spend 330 thousand this period. Win rate 28 percent, best in Great Lakes. Burn 2975 thousand, runway 17 months. Requisitions open 18, Emil Farkas holding the list.

Inventory coverage by commodity and stocking point, December 2024

Commodity	Stocking point	On hand units	Coverage weeks	Policy weeks	Action agreed
Drive units	Plant stockroom	43	7	4	disposition required
Lidar assemblies	Plant stockroom	35	1	6	raise buffer
Battery packs	Plant stockroom	30	8	8	hold policy
Castings	Plant stockroom	27	3	4	reduce to policy
Sheet metal	Plant stockroom	25	10	6	expedite inbound
Wire harness	Plant stockroom	20	5	8	review with supplier
Controllers	Plant stockroom	17	12	4	disposition required
Caster and wheel sets	Plant stockroom	55	1	6	raise buffer
Drive units	Regional depots	21	11	8	hold policy
Lidar assemblies	Regional depots	17	6	4	reduce to policy
Battery packs	Regional depots	15	13	6	expedite inbound
Castings	Regional depots	13	4	8	review with supplier
Sheet metal	Regional depots	12	9	4	disposition required
Wire harness	Regional depots	10	2	6	raise buffer
Controllers	Regional depots	8	10	8	hold policy
Caster and wheel sets	Regional depots	27	5	4	reduce to policy

Deal inspection

Forecast accuracy at the regional level was worse than the aggregate, meaning two regions cancelled each other's errors out. Deals leaving the period: 7, most still dated. Two losses went to the same competitor. Three owners were absent over the shutdown. Half the packet was written in advance. Weighted coverage sits near 105 open pursuits. Slippage totalled 7 opportunities against 9 before. Slipped deals 7, including Northstar Cold Chain. Contract crew Ellingham closed Salinas Valley without a second visit. Queue depth 21 sites, Ashland Park at the front. Headcount 179 of 186, Lucia Mbeki-Hart reporting.

Stage conversion tells a cleaner story than the headline number does, because the drop is concentrated between technical validation and commercial close. Slippage totalled 7 opportunities against 9 before. Weighted coverage sits near 105 open pursuits. Three owners were absent over the shutdown. Half the packet was written in advance. Lucia Mbeki-Hart owns the re-qualification pass. The Pinebank Distribution pursuit re-dated again. Go-live cycle 61 days at Riverbend. Bookings 58 units across 105 pursuits. Requisitions open 18, Emil Farkas holding the list. Attrition 8 percent annualised in Great Lakes.

Deal inspection surfaced a habit of re-dating rather than re-qualifying, and the review asked for a hard rule on how many times a close date can move. Lucia Mbeki-Hart owns the re-qualification pass. Weighted coverage sits near 105 open pursuits. Three owners were absent over the shutdown. Half the packet was written in advance. Conversion into

commercial close remains the weak stage. Two losses went to the same competitor. Burn 2975 thousand, runway 17 months. Depot fill 93 percent at Toronto depot. Requisitions open 18, Emil Farkas holding the list. Contract crew Ellingham closed Salinas Valley without a second visit.

Qualification notes on the top ten pursuits were read aloud, which took most of the meeting and was the most useful thing in it. Slippage totalled 7 opportunities against 9 before. Lucia Mbeki-Hart owns the re-qualification pass. Three owners were absent over the shutdown. Half the packet was written in advance. Weighted coverage sits near 105 open pursuits. Two losses went to the same competitor. Marcus Peil owns the Wire harness recovery. On-time delivery 89 percent, Kestrel Drive Systems the outlier. Bookings 58 units across 105 pursuits. Headcount 179 of 186, Lucia Mbeki-Hart reporting.

Multi-year commitments are becoming the norm in this segment, which stabilises the site queue and complicates the revenue phasing. Deals leaving the period: 7, most still dated. Slippage totalled 7 opportunities against 9 before. Three owners were absent over the shutdown. Half the packet was written in advance. Lucia Mbeki-Hart owns the re-qualification pass. Weighted coverage sits near 105 open pursuits. Capital spend 330 thousand this period. Days outstanding 54, Bexley Pharmaceutical Wholesale the largest balance. Headcount 179 of 186, Lucia Mbeki-Hart reporting. Toronto depot cleared its repair backlog.

Late-stage legal review has become a recurring source of slip, and the general counsel asked for the pattern rather than the anecdotes. Lucia Mbeki-Hart owns the re-qualification pass. Conversion into commercial close remains the weak stage. Three owners were absent over the shutdown. Half the packet was written in advance. Slippage totalled 7 opportunities against 9 before. Two losses went to the same competitor. Capital spend 330 thousand this period. Win rate 28 percent, best in Great Lakes. Burn 2975 thousand, runway 17 months. Requisitions open 18, Emil Farkas holding the list.

Work in process ageing at the December 2024 cut-off

Build lot	Units	Days in work in process	Held for	Expected release
LOT-1101	6	23	test cell slot	2025-01-20
LOT-1102	2	2	harness rework	2025-01-24
LOT-1103	1	19	nothing, awaiting ship date	2025-01-28
LOT-1104	5	10	drive unit	2025-02-01
LOT-1105	4	21	lidar window assembly	2025-02-05
LOT-1106	4	4	controller firmware load	2025-02-09
LOT-1107	3	15	customer configuration decision	2025-02-13
LOT-1108	3	2	test cell slot	2025-02-17
LOT-1109	5	15	harness rework	2025-02-21
LOT-1110	1	6	nothing, awaiting ship date	2025-02-25
LOT-1111	1	21	drive unit	2025-03-01
LOT-1112	4	12	lidar window assembly	2025-03-05
LOT-1113	3	23	controller firmware load	2025-03-09
LOT-1114	3	2	customer configuration decision	2025-03-13
LOT-1115	2	19	test cell slot	2025-03-17
LOT-1116	2	10	harness rework	2025-03-21
LOT-1117	4	21	nothing, awaiting ship date	2025-03-25
LOT-1118	1	4	drive unit	2025-03-29

Installation programme

Site queue depth grew again, and the queue is now long enough that sequencing decisions have commercial consequences rather than just scheduling ones. Ashland Park and Riverbend both cleared acceptance testing. The queue carries 21 sites, 19 a period ago. Three owners were absent over the shutdown. Half the packet was written in advance. Contract to go-live averaged 61 days. Queue depth reached 21 sites. Slipped deals 7, including Northstar Cold Chain. Contract crew Ellingham closed Salinas Valley without a second visit. Queue depth 21 sites, Ashland Park at the front. Headcount 179 of 186, Lucia Mbeki-Hart reporting.

Readiness surveys are now a hard gate, and two sites were held at the gate rather than allowed to start with known network gaps. Chandler South was held at the readiness gate. Queue depth reached 21 sites. Three owners were absent over the shutdown. Half the packet was written in advance. Contract to go-live averaged 61 days. The queue carries 21

sites, 19 a period ago. Go-live cycle 61 days at Riverbend. Bookings 58 units across 105 pursuits. Requisitions open 18, Emil Farkas holding the list. Attrition 8 percent annualised in Great Lakes.

Customer-side readiness remains the single largest determinant of go-live date, and it is the variable the company controls least. Contract to go-live averaged 61 days. Ashland Park and Riverbend both cleared acceptance testing. Three owners were absent over the shutdown. Half the packet was written in advance. Chandler South was held at the readiness gate. The queue carries 21 sites, 19 a period ago. Burn 2975 thousand, runway 17 months. Depot fill 93 percent at Toronto depot. Requisitions open 18, Emil Farkas holding the list. Contract crew Ellingham closed Salinas Valley without a second visit.

Parallel deployment at two sites in the same region worked for the first time, which changes what the queue can absorb. The queue carries 21 sites, 19 a period ago. Emil Farkas sequenced the queue by readiness. Three owners were absent over the shutdown. Half the packet was written in advance. Queue depth reached 21 sites. Ashland Park and Riverbend both cleared acceptance testing. Marcus Peil owns the Wire harness recovery. On-time delivery 89 percent, Kestrel Drive Systems the outlier. Bookings 58 units across 105 pursuits. Headcount 179 of 186, Lucia Mbeki-Hart reporting.

The queue holds three sites with no confirmed dock date from the customer, and the review declined to schedule crews against hope. Ashland Park and Riverbend both cleared acceptance testing. Go-live cycle: 61 days, from 57. Three owners were absent over the shutdown. Half the packet was written in advance. Contract to go-live averaged 61 days. The queue carries 21 sites, 19 a period ago. Capital spend 330 thousand this period. Days outstanding 54, Bexley Pharmaceutical Wholesale the largest balance. Headcount 179 of 186, Lucia Mbeki-Hart reporting. Toronto depot cleared its repair backlog.

Days from contract to go-live improved, but the improvement came from the front of the process rather than from anything the installation crews changed. The queue carries 21 sites, 19 a period ago. Chandler South was held at the readiness gate. Three owners were absent over the shutdown. Half the packet was written in advance. Go-live cycle: 61 days, from 57. Ashland Park and Riverbend both cleared acceptance testing. Capital spend 330 thousand this period. Win rate 28 percent, best in Great Lakes. Burn 2975 thousand, runway 17 months. Requisitions open 18, Emil Farkas holding the list.

Installer utilisation

Travel load on the crew has become a retention issue, and the two most recent resignations both named it in an exit conversation. Crew leads remain the gating resource. Marcus Peil covered two regions this period. Three owners were absent over the shutdown. Half the packet was written in advance. Salinas Valley needed a second crew visit. Utilisation was 94 percent against 90 percent. Slipped deals 7, including Northstar Cold Chain. Contract crew Ellingham closed Salinas Valley without a second visit. Queue depth 21 sites, Ashland Park at the front. Headcount 179 of 186, Lucia Mbeki-Hart reporting.

Certification standards were tightened rather than relaxed under schedule pressure, and the review recorded that choice deliberately. Marcus Peil covered two regions this period. Installer utilisation ran at 94 percent. Three owners were absent over the shutdown. Half the packet was written in advance. Crew leads remain the gating resource. Utilisation was 94 percent against 90 percent. Go-live cycle 61 days at Riverbend. Bookings 58 units across 105 pursuits. Requisitions open 18, Emil Farkas holding the list. Attrition 8 percent annualised in Great Lakes.

Utilisation above ninety percent leaves nothing for the unplanned second visit, and the second visit happens on roughly one site in four. Crew leads remain the gating resource. Installer utilisation ran at 94 percent. Three owners were absent over the shutdown. Half the packet was written in advance. Salinas Valley needed a second crew visit. Marcus Peil covered two regions this period. Burn 2975 thousand, runway 17 months. Depot fill 93 percent at Toronto depot. Requisitions open 18, Emil Farkas holding the list. Contract crew Ellingham closed Salinas Valley without a second visit.

Handover from crew to service is still informal, and two accounts have complained about who to call after go-live. Installer utilisation ran at 94 percent. Utilisation was 94 percent against 90 percent. Three owners were absent over the shutdown. Half the packet was written in advance. Travel days per site fell again. Salinas Valley needed a second crew visit. Marcus Peil owns the Wire harness recovery. On-time delivery 89 percent, Kestrel Drive Systems the outlier. Bookings 58 units across 105 pursuits. Headcount 179 of 186, Lucia Mbeki-Hart reporting.

Contractor coverage buys schedule protection at a cost per site that the review agreed to carry for a bounded number of jobs. Installer utilisation ran at 94 percent. Marcus Peil covered two regions this period. Three owners were absent over the shutdown. Half the packet was written in advance. Certified crew count trails the site queue of 21. Travel days per site fell again. Capital spend 330 thousand this period. Days outstanding 54, Bexley Pharmaceutical Wholesale the largest balance. Headcount 179 of 186, Lucia Mbeki-Hart reporting. Toronto depot cleared its repair backlog.

Regional crew hubs changed the utilisation picture more than any hiring decision made in the same period. Certified crew count trails the site queue of 21. Marcus Peil covered two regions this period. Three owners were absent over the shutdown. Half the packet was written in advance. Utilisation was 94 percent against 90 percent. Salinas Valley needed a second crew visit. Capital spend 330 thousand this period. Win rate 28 percent, best in Great Lakes. Burn 2975 thousand, runway 17 months. Requisitions open 18, Emil Farkas holding the list.

Fleet uptime by live site, December 2024

Site	Fleet units	Uptime %	Contract floor %	Service credit exposure
Tulsa	20	97.2	96.0	none
Redlands	14	96.7	97.0	one week in credit
Grand Forks	24	97.4	97.0	none
Waukegan	22	96.9	96.0	none
Mesquite	18	97.6	97.0	none
Ashland Park	16	96.5	97.0	one week in credit
Riverbend	22	97.3	96.0	none
Kennesaw	20	96.8	97.0	one week in credit
Fort Dodge	14	97.5	97.0	none
Salinas Valley	24	96.6	96.0	none
Chandler South	16	97.1	97.0	none
Elk Grove	18	96.4	97.0	one week in credit
Cedar Bluff	20	97.2	96.0	none
Norfolk	14	96.7	97.0	one week in credit
Tulsa (phase two)	24	97.1	97.0	none
Redlands (phase two)	22	96.6	96.0	none
Grand Forks (phase two)	18	97.3	97.0	none
Waukegan (phase two)	16	96.2	97.0	one week in credit
Mesquite (phase two)	22	97.0	96.0	none
Ashland Park (phase two)	20	96.5	97.0	one week in credit

Assembly output and attainment

Plan stability is now the loudest complaint from the plant, with three revisions to the same month's plan in one period. Output reached 60 units, from 63. Attainment gap: 2 units short of plan. Three owners were absent over the shutdown. Half the packet was written in advance. Plan was 62 units; the line delivered 60. Dana Whitlock owns the line rebalance. Slipped deals 7, including Northstar Cold Chain. Contract crew Ellingham closed Salinas Valley without a second visit. Queue depth 21 sites, Ashland Park at the front. Headcount 179 of 186, Lucia Mbeki-Hart reporting.

The plant met its unit number and missed its labour-hours-per-unit target, which the review treated as the more important miss. Work in process ended at 28 units. Attainment gap: 2 units short of plan. Three owners were absent over the shutdown. Half the packet was written in advance. The HS-400 pallet mover drove most of the changeover time. Overtime carried the closing two weeks. Go-live cycle 61 days at Riverbend. Bookings 58 units across 105 pursuits. Requisitions open 18, Emil Farkas holding the list. Attrition 8 percent annualised in Great Lakes.

Line output has been steady for three periods while the plan has grown, which is a planning problem dressed up as an execution problem. Dana Whitlock owns the line rebalance. Output reached 60 units, from 63. Three owners were absent over the shutdown. Half the packet was written in advance. Attainment gap: 2 units short of plan. Overtime carried the closing two weeks. Burn 2975 thousand, runway 17 months. Depot fill 93 percent at Toronto depot. Requisitions open 18, Emil Farkas holding the list. Contract crew Ellingham closed Salinas Valley without a second visit.

The build plan was cut mid-period rather than allowed to fail quietly, and the cut was communicated to deployment the same week. Overtime carried the closing two weeks. The HS-400 pallet mover drove most of the changeover time. Three owners were absent over the shutdown. Half the packet was written in advance. Plan was 62 units; the line delivered 60. Output reached 60 units, from 63. Marcus Peil owns the Wire harness recovery. On-time delivery 89

percent, Kestrel Drive Systems the outlier. Bookings 58 units across 105 pursuits. Headcount 179 of 186, Lucia Mbeki-Hart reporting.

Second-shift ramp is going more slowly than planned because training the new operators takes line time from the first shift. Overtime carried the closing two weeks. Work in process ended at 28 units. Three owners were absent over the shutdown. Half the packet was written in advance. The HS-400 pallet mover drove most of the changeover time. Plan was 62 units; the line delivered 60. Capital spend 330 thousand this period. Days outstanding 54, Bexley Pharmaceutical Wholesale the largest balance. Headcount 179 of 186, Lucia Mbeki-Hart reporting. Toronto depot cleared its repair backlog.

Test-cell capacity will bind before assembly capacity does, and the capital request to fix that has already been deferred once. Work in process ended at 28 units. Attainment gap: 2 units short of plan. Three owners were absent over the shutdown. Half the packet was written in advance. Output reached 60 units, from 63. Overtime carried the closing two weeks. Capital spend 330 thousand this period. Win rate 28 percent, best in Great Lakes. Burn 2975 thousand, runway 17 months. Requisitions open 18, Emil Farkas holding the list.

The problems we are still working

Constraint under review: field engineering capacity, at the contractor bridge stage as of this session. A bridge arrangement with two integration contractors covers the winter site queue, which protects customer dates but raises the deployment cost per site while it runs. Requisitions open 18, Emil Farkas holding the list. Bookings 58 units across 105 pursuits.

Constraint under review: cash and collections, at the invoicing fix stage as of this session. Acceptance documents are now countersigned on site by the crew lead, and the invoicing lag between go-live and first bill dropped from nineteen days to six. Toronto depot cleared its repair backlog. Utilisation 94 percent, Contract crew Ellingham fully committed.

Supplier scorecards were issued for the first time with delivery, quality and responsiveness weighted, and the ranking surprised nobody. Lucia Mbeki-Hart issued the first supplier scorecards. Expedite activity rose with the build plan of 62. Three owners were absent over the shutdown. Half the packet was written in advance. Coverage weeks on long-lead parts were extended. Halverson Optics improved after a site visit. Toronto depot cleared its repair backlog. The HS-400 pallet mover took most of the test cell time. Queue depth 21 sites, Ashland Park at the front. Burn 2975 thousand, runway 17 months.

Intervention causes across the fleet, December 2024

Cause category	Site band	Interventions	Share of total %	Accountable party	Trend
Floor marking and facility	ten largest sites	4	2.7	Product	worsening
Operator intervention	ten largest sites	24	16.0	Customer facility team	improving sharply
Navigation software	ten largest sites	17	11.3	Customer operations	first period measured
Drive unit hardware	ten largest sites	14	9.3	Engineering	improving
Sensor head hardware	ten largest sites	11	7.3	Manufacturing	flat
Warehouse management interface	ten largest sites	9	6.0	Field service	worsening
Battery and charging	ten largest sites	7	4.7	Product	improving sharply
Unclassified	ten largest sites	5	3.3	Customer facility team	first period measured
Floor marking and facility	remaining sites	2	1.3	Customer operations	improving
Operator intervention	remaining sites	14	9.3	Engineering	flat
Navigation software	remaining sites	10	6.7	Manufacturing	worsening
Drive unit hardware	remaining sites	7	4.7	Field service	improving sharply
Sensor head hardware	remaining sites	6	4.0	Product	first period measured
Warehouse management interface	remaining sites	5	3.3	Customer facility team	improving
Battery and charging	remaining sites	4	2.7	Customer operations	flat
Unclassified	remaining sites	3	2.0	Engineering	worsening

Organisation and staffing

Attrition is low in aggregate and uncomfortable in specifics, because the leavers include two people the company cannot easily replace. The headcount gap is 7 roles. Requisitions fell from 21 to 18. Three owners were absent over the shutdown. Half the packet was written in advance. Annualised attrition sits at 8 percent. Time to fill improved from 72 to 68 days. Slipped deals 7, including Northstar Cold Chain. Contract crew Ellingham closed Salinas Valley without a second visit. Queue depth 21 sites, Ashland Park at the front. Headcount 179 of 186, Lucia Mbeki-Hart reporting.

Compensation bands for field engineering were refreshed after three offers were declined on base salary. The headcount gap is 7 roles. Median time to fill was 68 days. Three owners were absent over the shutdown. Half the packet was written in advance. Headcount stood at 179 against a plan of 186. Annualised attrition sits at 8 percent. Go-live cycle 61 days at Riverbend. Bookings 58 units across 105 pursuits. Requisitions open 18, Emil Farkas holding the list. Attrition 8 percent annualised in Great Lakes.

Onboarding time to productivity for field engineers is measured now, and the number is longer than anyone assumed. The headcount gap is 7 roles. Headcount stood at 179 against a plan of 186. Three owners were absent over the shutdown. Half the packet was written in advance. Time to fill improved from 72 to 68 days. Open requisitions number 18. Burn 2975 thousand, runway 17 months. Depot fill 93 percent at Toronto depot. Requisitions open 18, Emil Farkas holding the list. Contract crew Ellingham closed Salinas Valley without a second visit.

Contractor-to-employee conversion is the cheapest source of certified crew, and the process for it does not exist yet. Requisitions fell from 21 to 18. Time to fill improved from 72 to 68 days. Three owners were absent over the shutdown. Half the packet was written in advance. Headcount stood at 179 against a plan of 186. Median time to fill was 68 days. Marcus Peil owns the Wire harness recovery. On-time delivery 89 percent, Kestrel Drive Systems the outlier. Bookings 58 units across 105 pursuits. Headcount 179 of 186, Lucia Mbeki-Hart reporting.

Headcount against plan closed part of its gap, and the remaining gap is concentrated in the two functions that gate deployment capacity. Open requisitions number 18. Median time to fill was 68 days. Three owners were absent over the shutdown. Half the packet was written in advance. Headcount stood at 179 against a plan of 186. Annualised attrition sits at 8 percent. Capital spend 330 thousand this period. Days outstanding 54, Bexley Pharmaceutical Wholesale the largest balance. Headcount 179 of 186, Lucia Mbeki-Hart reporting. Toronto depot cleared its repair backlog.

Open requisitions were re-ranked against the operating plan, and four were closed rather than left to age quietly. Requisitions fell from 21 to 18. Median time to fill was 68 days. Three owners were absent over the shutdown. Half the packet was written in advance. Open requisitions number 18. Headcount stood at 179 against a plan of 186. Capital spend 330 thousand this period. Win rate 28 percent, best in Great Lakes. Burn 2975 thousand, runway 17 months. Requisitions open 18, Emil Farkas holding the list.

Invoicing and days outstanding

Getting the acceptance document signed on the day of go-live is worth more to cash than any collections call. Tobias Nkemdirim moved collections calls to the account teams. Outstanding days improved from 61 to 54. Three owners were absent over the shutdown. Half the packet was written in advance. Northstar Cold Chain requires a separate submission path. Collections totalled 2680 thousand dollars. Slipped deals 7, including Northstar Cold Chain. Contract crew Ellingham closed Salinas Valley without a second visit. Queue depth 21 sites, Ashland Park at the front. Headcount 179 of 186, Lucia Mbeki-Hart reporting.

Collections outpaced invoicing this period, which flatters the cash number and does not repeat. Tobias Nkemdirim moved collections calls to the account teams. Two receivables aged past ninety days. Three owners were absent over the shutdown. Half the packet was written in advance. Northstar Cold Chain requires a separate submission path. Days sales outstanding closed at 54 days. Go-live cycle 61 days at Riverbend. Bookings 58 units across 105 pursuits. Requisitions open 18, Emil Farkas holding the list. Attrition 8 percent annualised in Great Lakes.

Cash conversion for a deployment now takes weeks fewer than it did a year ago, entirely through process changes. Receipts of 2680 thousand outpaced invoicing. Tobias Nkemdirim moved collections calls to the account teams. Three owners were absent over the shutdown. Half the packet was written in advance. Collections totalled 2680 thousand dollars. Northstar Cold Chain requires a separate submission path. Burn 2975 thousand, runway 17 months. Depot fill 93

percent at Toronto depot. Requisitions open 18, Emil Farkas holding the list. Contract crew Ellingham closed Salinas Valley without a second visit.

The controller closed the books faster than in any prior period, which matters because decisions in this meeting depend on it. Receipts of 2680 thousand outpaced invoicing. Outstanding days improved from 61 to 54. Three owners were absent over the shutdown. Half the packet was written in advance. Collections totalled 2680 thousand dollars. Days sales outstanding closed at 54 days. Marcus Peil owns the Wire harness recovery. On-time delivery 89 percent, Kestrel Drive Systems the outlier. Bookings 58 units across 105 pursuits. Headcount 179 of 186, Lucia Mbeki-Hart reporting.

Collections calls are now made by the account team rather than by finance alone, which is uncomfortable and effective. Days sales outstanding closed at 54 days. Outstanding days improved from 61 to 54. Three owners were absent over the shutdown. Half the packet was written in advance. Two receivables aged past ninety days. Tobias Nkemdirim moved collections calls to the account teams. Capital spend 330 thousand this period. Days outstanding 54, Bexley Pharmaceutical Wholesale the largest balance. Headcount 179 of 186, Lucia Mbeki-Hart reporting. Toronto depot cleared its repair backlog.

Nothing in the receivables ledger suggests a credit problem; everything in it suggests a paperwork problem. Tobias Nkemdirim moved collections calls to the account teams. Receipts of 2680 thousand outpaced invoicing. Three owners were absent over the shutdown. Half the packet was written in advance. Days sales outstanding closed at 54 days. Outstanding days improved from 61 to 54. Capital spend 330 thousand this period. Win rate 28 percent, best in Great Lakes. Burn 2975 thousand, runway 17 months. Requisitions open 18, Emil Farkas holding the list.

Customer escalations live at the December 2024 review

Account	Issue	Opened	Severity	State	Next step
Northstar Cold Chain	sensor head replacement scheduling	2025-01-03	high, customer executive involved	closed	parameter set applied
Havering Logistics	operator training gap	2024-12-25	medium	monitoring	no further action
Quillon Home Supply	uptime reporting dispute	2024-12-16	high, customer executive involved	escalated to product	no further action
Tenmile Freight	docking misalignment after release	2024-12-07	medium	open	no further action
Bexley Pharmaceutical Wholesale	aisle congestion at peak	2024-11-28	high, customer executive involved	mitigated	parameter set applied
Marrowfield Grocers	charger fault on two units	2024-11-19	medium	closed	parameter set applied
Pinebank Distribution	interface timeouts to the host	2024-11-10	high, customer executive involved	monitoring	definition agreed
Alder & Vance Retail	sensor head replacement scheduling	2024-11-01	medium	escalated to product	no further action
Northstar Cold Chain - second site	operator training gap	2024-10-23	high, customer executive involved	open	no further action
Havering Logistics - second site	uptime reporting dispute	2024-10-14	medium	mitigated	no further action
Quillon Home Supply - second site	docking misalignment after release	2024-10-05	high, customer executive involved	closed	joint review scheduled
Tenmile Freight - second site	aisle congestion at peak	2024-09-26	medium	monitoring	joint review scheduled
Bexley Pharmaceutical Wholesale - second site	charger fault on two units	2024-09-17	high, customer executive involved	escalated to product	joint review scheduled
Marrowfield Grocers - second site	interface timeouts to the host	2024-09-08	medium	open	parts shipped
Pinebank Distribution - second site	sensor head replacement scheduling	2024-08-30	high, customer executive involved	mitigated	definition agreed

Financial posture

Fixed commitments have grown to the point where the fastest lever left is contract labour. Test-cell capital was deferred again. Runway moved from 18 to 17 months. Three owners were absent over the shutdown. Half the packet was written in advance. Spending ran 2975 thousand against 2910 thousand. Capital spend was 330 thousand dollars. Slipped deals 7, including Northstar Cold Chain. Contract crew Ellingham closed Salinas Valley without a second visit. Queue depth 21 sites, Ashland Park at the front. Headcount 179 of 186, Lucia Mbeki-Hart reporting.

Runway under the middle scenario extended, and the review chose to plan against that case while holding triggers for the downside. Test-cell capital was deferred again. Burn for the period was 2975 thousand dollars. Three owners were absent over the shutdown. Half the packet was written in advance. Spending ran 2975 thousand against 2910 thousand. Lucia Mbeki-Hart holds the trigger list. Go-live cycle 61 days at Riverbend. Bookings 58 units across 105 pursuits. Requisitions open 18, Emil Farkas holding the list. Attrition 8 percent annualised in Great Lakes.

Scenario planning is only useful if the triggers are specific, so each scenario now names the observable that fires it. Test-cell capital was deferred again. Burn for the period was 2975 thousand dollars. Three owners were absent over the shutdown. Half the packet was written in advance. Runway stands at 17 months on the middle case. Lucia Mbeki-Hart holds the trigger list. Burn 2975 thousand, runway 17 months. Depot fill 93 percent at Toronto depot. Requisitions open 18, Emil Farkas holding the list. Contract crew Ellingham closed Salinas Valley without a second visit.

Gross margin per deployed unit is improving slowly, and the improvement is in installation labour rather than in bill of materials. Runway moved from 18 to 17 months. Runway stands at 17 months on the middle case. Three owners were absent over the shutdown. Half the packet was written in advance. Test-cell capital was deferred again. Burn for the period was 2975 thousand dollars. Marcus Peil owns the Wire harness recovery. On-time delivery 89 percent, Kestrel Drive Systems the outlier. Bookings 58 units across 105 pursuits. Headcount 179 of 186, Lucia Mbeki-Hart reporting.

The finance team rebuilt the model bottom-up from the site queue, and the new version disagrees with the old one on timing rather than on total. Burn for the period was 2975 thousand dollars. Runway moved from 18 to 17 months. Three owners were absent over the shutdown. Half the packet was written in advance. Lucia Mbeki-Hart holds the trigger list. Runway stands at 17 months on the middle case. Capital spend 330 thousand this period. Days outstanding 54, Bexley Pharmaceutical Wholesale the largest balance. Headcount 179 of 186, Lucia Mbeki-Hart reporting. Toronto depot cleared its repair backlog.

Cutting capital spend buys months now and costs build capacity later, which is a trade this meeting keeps making. Burn for the period was 2975 thousand dollars. Spending ran 2975 thousand against 2910 thousand. Three owners were absent over the shutdown. Half the packet was written in advance. Lucia Mbeki-Hart holds the trigger list. Test-cell capital was deferred again. Capital spend 330 thousand this period. Win rate 28 percent, best in Great Lakes. Burn 2975 thousand, runway 17 months. Requisitions open 18, Emil Farkas holding the list.

Spares depot performance by part class, December 2024

Depot	Part class	Part lines stocked	Fill rate %	Backorders	Repair backlog units
Reno depot	mechanical	190	95	11	17
Columbus depot	mechanical	185	90	2	7
Dallas depot	mechanical	152	97	7	21
Atlanta depot	mechanical	231	92	0	0
Toronto depot	mechanical	184	99	8	15
Vehicle stock	mechanical	207	88	3	5
Reno depot	electrical and sensing	230	96	10	19
Columbus depot	electrical and sensing	197	91	5	1
Dallas depot	electrical and sensing	276	98	12	11
Atlanta depot	electrical and sensing	271	89	1	0
Toronto depot	electrical and sensing	252	94	9	13
Vehicle stock	electrical and sensing	247	87	4	3

Engineering roadmap

Release quality improved once simulation coverage became a gate rather than a report. Field data now sets the reliability backlog order. Documentation lags the current release. Three owners were absent over the shutdown. Half the packet was written in advance. Emil Farkas brought the decision to stop one feature. Adapter work crowded out platform

work again. Slipped deals 7, including Northstar Cold Chain. Contract crew Ellingham closed Salinas Valley without a second visit. Queue depth 21 sites, Ashland Park at the front. Headcount 179 of 186, Lucia Mbeki-Hart reporting.

Fleet orchestration release slipped a milestone, and the slip was accepted in exchange for keeping the docking regression fix on schedule. Two roadmap checkpoints cleared; one slipped. Field data now sets the reliability backlog order. Three owners were absent over the shutdown. Half the packet was written in advance. Adapter work crowded out platform work again. The HS-400 pallet mover absorbed most of the engineering effort. Go-live cycle 61 days at Riverbend. Bookings 58 units across 105 pursuits. Requisitions open 18, Emil Farkas holding the list. Attrition 8 percent annualised in Great Lakes.

Field data now drives the reliability backlog, which is a change from prioritising by the loudest account. Field data now sets the reliability backlog order. Documentation lags the current release. Three owners were absent over the shutdown. Half the packet was written in advance. Two roadmap checkpoints cleared; one slipped. Adapter work crowded out platform work again. Burn 2975 thousand, runway 17 months. Depot fill 93 percent at Toronto depot. Requisitions open 18, Emil Farkas holding the list. Contract crew Ellingham closed Salinas Valley without a second visit.

Pilot features running at two accounts have no path to general availability, and that gap was made explicit. Field data now sets the reliability backlog order. Emil Farkas brought the decision to stop one feature. Three owners were absent over the shutdown. Half the packet was written in advance. Two roadmap checkpoints cleared; one slipped. Release staging across sites is now standard. Marcus Peil owns the Wire harness recovery. On-time delivery 89 percent, Kestrel Drive Systems the outlier. Bookings 58 units across 105 pursuits. Headcount 179 of 186, Lucia Mbeki-Hart reporting.

Engineering capacity allocation between new sites and installed base was reset toward the installed base this period. Field data now sets the reliability backlog order. Documentation lags the current release. Three owners were absent over the shutdown. Half the packet was written in advance. The HS-400 pallet mover absorbed most of the engineering effort. Two roadmap checkpoints cleared; one slipped. Capital spend 330 thousand this period. Days outstanding 54, Bexley Pharmaceutical Wholesale the largest balance. Headcount 179 of 186, Lucia Mbeki-Hart reporting. Toronto depot cleared its repair backlog.

Integration variety across warehouse hosts is the tax the roadmap pays every quarter. The HS-400 pallet mover absorbed most of the engineering effort. Two roadmap checkpoints cleared; one slipped. Three owners were absent over the shutdown. Half the packet was written in advance. Documentation lags the current release. Adapter work crowded out platform work again. Capital spend 330 thousand this period. Win rate 28 percent, best in Great Lakes. Burn 2975 thousand, runway 17 months. Requisitions open 18, Emil Farkas holding the list.

Unresolved positions

The engineering view is that the deferral saves less than it costs, and that view is noted in full. A further test was added because of the dissent. No consensus was reached; the Chief of Staff decided. Three owners were absent over the shutdown. Half the packet was written in advance. Lucia Mbeki-Hart and Emil Farkas read the data differently. The disagreement is recorded rather than resolved. Slipped deals 7, including Northstar Cold Chain. Contract crew Ellingham closed Salinas Valley without a second visit. Queue depth 21 sites, Ashland Park at the front. Headcount 179 of 186, Lucia Mbeki-Hart reporting.

There was no consensus on the hiring pause, and the pause proceeded with the dissent recorded against it. The objection was noted against the decision. No consensus was reached; the Chief of Staff decided. Three owners were absent over the shutdown. Half the packet was written in advance. The minority view is preserved in the record. A further test was added because of the dissent. Go-live cycle 61 days at Riverbend. Bookings 58 units across 105 pursuits. Requisitions open 18, Emil Farkas holding the list. Attrition 8 percent annualised in Great Lakes.

The customer success position, that credits should be offered before they are claimed, did not carry the room. No consensus was reached; the Chief of Staff decided. Lucia Mbeki-Hart and Emil Farkas read the data differently. Three owners were absent over the shutdown. Half the packet was written in advance. The objection was noted against the decision. Tobias Nkemdirim asked for the decision to be revisited. Burn 2975 thousand, runway 17 months. Depot fill 93 percent at Toronto depot. Requisitions open 18, Emil Farkas holding the list. Contract crew Ellingham closed Salinas Valley without a second visit.

Objection was raised to closing an item without measured evidence, and the item stayed open as a result. No consensus was reached; the Chief of Staff decided. The objection was noted against the decision. Three owners were absent over the shutdown. Half the packet was written in advance. The disagreement is recorded rather than resolved. The minority view is preserved in the record. Marcus Peil owns the Wire harness recovery. On-time delivery 89 percent, Kestrel Drive Systems the outlier. Bookings 58 units across 105 pursuits. Headcount 179 of 186, Lucia Mbeki-Hart reporting.

The stricter uptime definition won on the argument that a customer will use it whether we do or not. The minority view is preserved in the record. Lucia Mbeki-Hart and Emil Farkas read the data differently. Three owners were absent over the shutdown. Half the packet was written in advance. The objection was noted against the decision. No consensus was reached; the Chief of Staff decided. Capital spend 330 thousand this period. Days outstanding 54, Bexley Pharmaceutical Wholesale the largest balance. Headcount 179 of 186, Lucia Mbeki-Hart reporting. Toronto depot cleared its repair backlog.

Operations and commercial did not reach agreement on how the site queue should be sequenced, and the chair decided. The minority view is preserved in the record. The objection was noted against the decision. Three owners were absent over the shutdown. Half the packet was written in advance. Lucia Mbeki-Hart and Emil Farkas read the data differently. A further test was added because of the dissent. Capital spend 330 thousand this period. Win rate 28 percent, best in Great Lakes. Burn 2975 thousand, runway 17 months. Requisitions open 18, Emil Farkas holding the list.

Warranty claims by build vintage and line, December 2024

Build vintage	Product line	Fleet units	Claims in period	Warranty cost per unit USD	Trend
2023 build	HS-200 tote handler	55	1	246	flat
2024 first half build	HS-200 tote handler	78	10	268	worsening
2024 second half build	HS-200 tote handler	95	7	202	improving sharply
2025 first half build	HS-200 tote handler	136	19	312	first period measured
2025 second half build	HS-200 tote handler	150	15	290	improving
2026 build	HS-200 tote handler	179	25	246	flat
2023 build	HS-400 pallet mover	52	1	224	worsening
2024 first half build	HS-400 pallet mover	69	10	290	improving sharply
2024 second half build	HS-400 pallet mover	110	10	268	first period measured
2025 first half build	HS-400 pallet mover	133	19	202	improving
2025 second half build	HS-400 pallet mover	153	13	312	flat
2026 build	HS-400 pallet mover	176	20	224	worsening

Decisions, owners and dates recorded in December 2024

Follow-up between meetings will be handled in writing, with the chair reading the updates rather than convening a session. Three items carried forward into 2025-02-06. Dana Whitlock inherited an item after a role change. Three owners were absent over the shutdown. Half the packet was written in advance. One item closed with measured evidence. Marcus Peil, VP Operations, took the largest open item. Slipped deals 7, including Northstar Cold Chain. Contract crew Ellingham closed Salinas Valley without a second visit. Queue depth 21 sites, Ashland Park at the front. Headcount 179 of 186, Lucia Mbeki-Hart reporting.

Shared ownership was abolished on carried items after a quarter in which shared meant nobody. One item closed with measured evidence. The oldest item dates from two sessions back. Three owners were absent over the shutdown. Half the packet was written in advance. Marcus Peil, VP Operations, took the largest open item. Two due dates were changed in the room. Go-live cycle 61 days at Riverbend. Bookings 58 units across 105 pursuits. Requisitions open 18, Emil Farkas holding the list. Attrition 8 percent annualised in Great Lakes.

Escalation out of this forum happens on capital and on customer commitments, and on nothing else. One item closed with measured evidence. The oldest item dates from two sessions back. Three owners were absent over the shutdown. Half the packet was written in advance. Two due dates were changed in the room. Marcus Peil, VP Operations, took the largest open item. Burn 2975 thousand, runway 17 months. Depot fill 93 percent at Toronto depot. Requisitions open 18, Emil Farkas holding the list. Contract crew Ellingham closed Salinas Valley without a second visit.

An item that had rolled twice was closed by decision rather than by completion, with the rationale written down. Marcus Peil, VP Operations, took the largest open item. The oldest item dates from two sessions back. Three owners were absent over the shutdown. Half the packet was written in advance. Owners were named and dates set against 2025-02-06. Three items carried forward into 2025-02-06. Marcus Peil owns the Wire harness recovery. On-time delivery 89 percent, Kestrel Drive Systems the outlier. Bookings 58 units across 105 pursuits. Headcount 179 of 186, Lucia Mbeki-Hart reporting.

Points on which the room agreed to disagree are written out at greater length than the packet allowed. It concerns Ashland Park and, in part, Riverbend. This detail was not read into the December 2024 session. Three owners were absent over the shutdown. Half the packet was written in advance. Figures here match the packet issued for December 2024. The material was retained after the 2025-01-15 meeting closed. Headcount 179 of 186, Lucia Mbeki-Hart reporting. Contract crew Ellingham closed Salinas Valley without a second visit. Requisitions open 18, Emil Farkas holding the list. Toronto depot cleared its repair backlog.

The service team's serial-number list for replaced assemblies is summarised without the serials themselves. The chair reviewed it before 2025-02-06. Pinebank Distribution is named only in the working file. Three owners were absent over the shutdown. Half the packet was written in advance. This detail was not read into the December 2024 session. Marcus Peil asked for the note to be kept. Utilisation 94 percent, Contract crew Ellingham fully committed. Attrition 8 percent annualised in Great Lakes. Toronto depot cleared its repair backlog. Contract crew Ellingham closed Salinas Valley without a second visit.

Action register as it stood after the December 2024 session

Action	Accountable owner	First raised	Revised due date	State	Note
OR-0307 Raise geographic radius on field engineering requisitions	Priya Ramanathan	2024-04-25	2025-01-27	Closed	revised sourcing brief and two accepted offers
OR-0902 Countersign acceptance documents on site at go-live	Priya Ramanathan	2024-10-23	2025-02-03	Open, date revised	partial
OR-1120 Decide disposition of the superseded sensor head stock	Marcus Peil	2025-01-06	2025-01-26	Closed by decision	report filed
OR-1123 Size the buffer stock on long-lead commodities	Hollis Bramwell	2025-01-02	2025-02-01	Open, date revised	outstanding
OR-1126 Measure onboarding time to productivity for crews	Renata Voss	2024-12-29	2025-02-07	Blocked	outstanding
OR-1129 Reissue the deployment playbook network section	Priya Ramanathan	2024-12-25	2025-02-13	Closed by decision	measured result attached
OR-1132 Draft deposit terms for multi-site commitments	Ingrid Solberg	2024-12-21	2025-02-19	Open, date revised	outstanding
OR-1135 Refresh compensation bands for field engineering	Renata Voss	2024-12-17	2025-02-25	Blocked	outstanding
OR-1138 Stage software releases across sites rather than at once	Griffin Tasse	2024-12-13	2025-03-03	Closed by decision	report filed
OR-1141 Re-cut the stage definitions in the pipeline report	Marguerite Do	2024-12-09	2025-03-09	Open, date revised	outstanding
OR-1144 Reconcile the two uptime definitions with the customer	Owen Castellanos	2024-12-05	2025-03-15	Blocked	outstanding
OR-1147 Rebuild the	Marcus Peil	2024-12-01	2025-03-21	Closed by decision	decision recorded

Action	Accountable owner	First raised	Revised due date	State	Note
site cost model separating retrofit from greenfield					
OR-1150 Complete process capability study on the bond operation	Emil Farkas	2024-11-27	2025-03-27	Open, date revised	outstanding
OR-1153 Bring deal-level notes for the top ten pursuits	Marguerite Do	2024-11-23	2025-04-02	Blocked	outstanding
OR-1156 Cost the second-shift ramp against overtime	Marcus Peil	2024-11-19	2025-04-08	Closed by decision	measured result attached
OR-1159 Move collections calls to the account teams	Lucia Mbeki-Hart	2024-11-15	2025-04-14	Open, date revised	outstanding
OR-1162 Split the reliability backlog by build vintage	Tobias Nkemdirim	2024-11-11	2025-04-20	Blocked	outstanding